Evaluating an Ensemble Score Filter Against the LETKF in the SPEEDY Atmospheric General Circulation Model

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Contents

List of Figures

List of Tables

Introduction

Numerical weather prediction depends on an accurate estimate of the atmospheric state. A forecast model advances that estimate in time, but uncertainty in the initial condition and imperfect knowledge of the evolving atmosphere cause the model trajectory to depart from the true state. Data assimilation (DA) reduces this uncertainty by combining a model forecast with observations and their estimated errors. Because atmospheric models have very large state dimensions, modern DA systems often use an ensemble of forecasts to represent uncertainty and estimate flow-dependent relationships among variables [4, 7].

The Local Ensemble Transform Kalman Filter (LETKF) is a well-established ensemble DA method. It performs a deterministic Kalman update in ensemble space within local spatial neighborhoods, making the analysis computationally tractable and reducing the influence of spurious long-range correlations. LETKF systems have been successfully applied to atmospheric general circulation models, including the NCEP global model [6, 12]. Under linear observation operators and approximately Gaussian forecast uncertainty, the Kalman analysis is strongly motivated by theory. In practice, multiplicative inflation and localization are required to compensate for finite-ensemble sampling error and insufficient ensemble spread.

The Gaussian and linear assumptions underlying ensemble Kalman updates can become restrictive when the prior ensemble is skewed or multimodal, or when the observation is a nonlinear function of the state. Non-Gaussian ensemble methods are therefore of considerable interest [1]. Score-based diffusion methods provide one recent approach. Rather than representing an analysis update only through a mean and covariance, a score filter uses the gradient of the log probability density, called the score, to guide samples through a reverse diffusion process. Bao et al. [2, 3] developed score-based filters for nonlinear and high-dimensional filtering, including a training-free Ensemble Score Filter (EnSF). Subsequent atmospheric experiments compared an EnSF with LETKF under observation networks of different density and nonlinearity [13]. These studies established the promise of score-based filtering, but much of its development has relied on low-dimensional benchmarks, such as Lorenz systems, or otherwise idealized geophysical models. Such settings are essential for method development but do not reproduce the multivariable, vertically resolved coupling of an atmospheric general circulation model.

The goal of this project is therefore to test whether the advantages of score-based assimilation persist in a more realistic atmospheric setting. The Simplified Parameterizations, Primitive Equation Dynamics model (SPEEDY) is an intermediate-complexity atmospheric general circulation model (AGCM). SPEEDY resolves global atmospheric circulation while remaining inexpensive enough for controlled ensemble experiments [8, 9]. The experiments use the hybrid Fortran 77/Fortran 90 SPEEDY implementation integrated with AMLCS-DA. This code base retains legacy Fortran 77 components alongside Fortran 90 routines. The model provides a useful bridge between low-dimensional test systems and the substantially greater cost and complexity of operational numerical weather prediction.

Both filters were implemented in the AMLCS-DA framework of Nino-Ruiz and Consuegra [10]. The existing LETKF implementation was modified to assimilate nonlinear wind speed and wind direction observations. The score filter was implemented in `sequential_methods.py` by adapting an EnSF prototype provided by Zixiang Xiong. Only LETKF and the score filter are considered in this project; other DA methods available in AMLCS-DA are outside its scope.

Four observing-system simulation experiments are evaluated. The first directly observes every state variable with a linear operator. The second applies a componentwise arctangent operator to all observations. The third replaces direct observations of the zonal and meridional wind components with wind speed and wind direction while retaining direct observations of temperature, humidity, and surface pressure. The final experiment observes only surface pressure. Together these cases address four important questions: how the methods behave in a favorable linear setting, how they respond to controlled nonlinearity, how they handle a physically interpretable nonlinear wind operator, and how effectively they use cross-variable information when only a small portion of the model state is observed.

Methods and Data

1 Sequential Data Assimilation

Let $\boldsymbol{x}_k$ denote the atmospheric state at assimilation cycle k and let $\boldsymbol{y}_k$ denote the observations. A forecast model $\mathcal{M}$ advances each analysis ensemble member from cycle $k-1$ to the background at cycle k:

$$\boldsymbol{x}_k^{b,(n)} = \mathcal{M}\left(\boldsymbol{x}_{k-1}^{a,(n)}\right), \qquad n = 1, \ldots, N_e, \tag{1}$$

where N_e is the ensemble size. The analysis step applies the observations through an observation operator $\mathcal{H}$ and an assumed observation-error distribution. In Bayesian form,

$$p(\boldsymbol{x}_k \mid \boldsymbol{y}_{1:k}) \propto p(\boldsymbol{y}_k \mid \boldsymbol{x}_k)\, p(\boldsymbol{x}_k \mid \boldsymbol{y}_{1:k-1}). \tag{2}$$

The LETKF and score filter use the same background ensemble and observations, but approximate and sample the posterior in different ways.

2 SPEEDY and AMLCS-DA

SPEEDY is a global spectral AGCM with simplified physical parameterizations. The formulation retains the large-scale primitive-equation dynamics and a reduced representation of processes such as convection, condensation, radiation, surface fluxes, and vertical diffusion [9]. The hybrid Fortran 77/Fortran 90 model used in this project is coupled to the AMLCS-DA workflow and produces NetCDF output suitable for ensemble experiments [10]. This hybrid implementation retains both legacy and modern Fortran components and is distinct from separate modern-Fortran rewrites of SPEEDY. The model is used at T21 horizontal resolution, corresponding to a 32×64 latitude–longitude grid.

The assimilated state contains zonal wind (UG1), meridional wind (VG1), temperature (TG1), specific humidity (TRG1), and surface pressure (PSG1). Wind and temperature have eight vertical levels. Humidity is stored on six levels, and surface pressure has a single level. For the three-dimensional atmospheric variables, level indices increase downward: level 0 is the highest model level and level 7 is the lowest, nearest-surface level. PSG1 uses a separate single-level surface field indexed as level 0. Table 1 summarizes the variables used in the plots. Humidity is converted from $\mathrm{kg\,kg^{-1}}$ to $\mathrm{g\,kg^{-1}}$ for interpretation. The pressure variable is the nondimensional logarithmic surface-pressure state used by SPEEDY.

Table 1: SPEEDY state variables evaluated in the experiments.

AMLCS name	Physical quantity	Units	Vertical treatment
UG1	Zonal wind	$\mathrm{m\,s^{-1}}$	Average of 8 levels
VG1	Meridional wind	$\mathrm{m\,s^{-1}}$	Average of 8 levels
TG1	Temperature	K	Average of 8 levels
TRG1	Specific humidity	$\mathrm{g\,kg^{-1}}$	Average of 6 levels
PSG1	Log surface pressure	nondimensional	Level 0

AMLCS-DA provides the forecast-analysis loop, experiment configuration, observation-network construction, state blocking, model execution, and error diagnostics [10]. The project retained this infrastructure and modified only the DA and observation components required for the two methods and four experiments. The same synthetic truth, initial ensemble, observation locations, ensemble size, and assimilation schedule were used for both methods within each experiment.

3 Local Ensemble Transform Kalman Filter

For a local state neighborhood, collect the background ensemble members in the matrix

$$\boldsymbol{X}^b = \left[\boldsymbol{x}^{b,(1)}, \dots, \boldsymbol{x}^{b,(N_e)}\right]. \tag{3}$$

Let $\overline{\boldsymbol{x}}^b$ be the ensemble mean and let $\boldsymbol{X}'^b$ contain the perturbations from that mean. The background ensemble is mapped into observation space, producing the observation-space mean $\overline{\boldsymbol{y}}^b$ and perturbation matrix $\boldsymbol{Y}'^b$. For a local observation-error covariance $\boldsymbol{R}$ and multiplicative inflation ρ, the analysis covariance in ensemble space is

$$\widetilde{\boldsymbol{P}}^a = \left[\frac{N_e - 1}{\rho}\boldsymbol{I} + (\boldsymbol{Y}'^b)^{\mathrm{T}}\boldsymbol{R}^{-1}\boldsymbol{Y}'^b\right]^{-1}. \tag{4}$$

The mean analysis weight is

$$\overline{\boldsymbol{w}}^a = \widetilde{\boldsymbol{P}}^a(\boldsymbol{Y}'^b)^{\mathrm{T}}\boldsymbol{R}^{-1}\left(\boldsymbol{y} - \overline{\boldsymbol{y}}^b\right), \tag{5}$$

and a symmetric square root of the analysis covariance supplies the ensemble transform. The resulting local analysis ensemble is

$$\boldsymbol{X}^a = \overline{\boldsymbol{x}}^b\boldsymbol{1}^{\mathrm{T}} + \boldsymbol{X}'^b\left[\overline{\boldsymbol{w}}^a\boldsymbol{1}^{\mathrm{T}} + \left((N_e - 1)\widetilde{\boldsymbol{P}}^a\right)^{1/2}\right]. \tag{6}$$

The update is repeated for each grid component using observations inside a local box. The localization radius r is measured in grid points on the 32×64 SPEEDY grid. Thus, r controls the local region from which observations and ensemble correlations influence a component.

The formulation follows the LETKF method described by Hunt et al. [6] and the global atmospheric implementation of Szunyogh et al. [12].

For nonlinear observations, the background ensemble is first evaluated through the nonlinear operator. The mean and perturbations are then computed in observation space and supplied to the same local ensemble-space update. This approach does not linearize the operator explicitly, but the analysis update itself remains an affine transformation of the background ensemble.

4 Ensemble Score Filter

The implemented score filter follows the reverse stochastic differential equation structure of the training-free EnSF [3]. It introduces a pseudotime $t \in [0, 1]$ and progressively connects the background distribution to a simple noisy distribution. In the implemented configuration, the conditional perturbation schedule is

$$\alpha_t = 1 - (1 - \epsilon_\alpha)t, \tag{7}$$

$$\beta_t^2 = \epsilon_\beta + (1 - \epsilon_\beta)t, \tag{8}$$

with $\epsilon_\alpha = 0.05$ and $\epsilon_\beta = 0.025$. The corresponding linear drift and diffusion coefficients are

$$f(t) = \frac{d \log \alpha_t}{dt} = -\frac{1 - \epsilon_\alpha}{\alpha_t}, \tag{9}$$

$$g(t)^2 = \frac{d\beta_t^2}{dt} - 2f(t)\beta_t^2. \tag{10}$$

At each pseudotime, the posterior score is decomposed into a background-prior term and an observation-likelihood term:

$$\boldsymbol{s}_{\text{post}}(\boldsymbol{x}_t, t) = \boldsymbol{s}_{\text{prior}}(\boldsymbol{x}_t, t) + \tau(t)\boldsymbol{s}_{\text{like}}(\boldsymbol{x}_t), \qquad \tau(t) = 1 - t. \tag{11}$$

For the paired ensemble approximation used here, the prior score for a background sample $\boldsymbol{x}^{b,(n)}$ is

$$\boldsymbol{s}_{\text{prior}}^{(n)}(\boldsymbol{x}_t, t) = -\frac{\boldsymbol{x}_t - \alpha_t \boldsymbol{x}^{b,(n)}}{\beta_t^2}. \tag{12}$$

For Gaussian observation error and a differentiable observation operator $\mathcal{H}$, the likelihood contribution is

$$\boldsymbol{s}_{\text{like}}(\boldsymbol{x}) = -\boldsymbol{J}_{\mathcal{H}}(\boldsymbol{x})^{\mathrm{T}} \boldsymbol{R}^{-1} \left[\mathcal{H}(\boldsymbol{x}) - \boldsymbol{y}\right], \tag{13}$$

where $\boldsymbol{J}_{\mathcal{H}}$ is the Jacobian of the observation operator. For a direct linear observation, Equation (13) reduces to the scaled residual between the pseudotime state and the observation. For the arctangent and wind experiments, the exact chain-rule derivatives are used.

Starting from standardized Gaussian samples at $t = 1$, the code integrates the reverse process toward $t = 0$ with an Euler-Maruyama update of the form

$$\boldsymbol{x}_{t-\Delta t} = \boldsymbol{x}_t + \Delta t \left[-f(t)\boldsymbol{x}_t + g(t)^2 \boldsymbol{s}_{\text{post}}(\boldsymbol{x}_t, t)\right] + \sqrt{\Delta t}\, g(t)\boldsymbol{z}, \tag{14}$$

where $\boldsymbol{z}$ is standard Gaussian noise. The experiments used 1,000 pseudotime steps, a fixed random seed of 42, and normalization based on the background ensemble spread. The resulting samples replace the observed components of the background ensemble and are rescaled before the forecast continues.

Unlike LETKF, the original EnSF update does not use a localized forecast-error covariance to spread an observed-component correction into unobserved state components. This limitation is important for interpreting the pressure-only experiment. Recent EnSF work introduces a regression step for this purpose [14], but that extension is not used here because the project evaluates the original score-filter formulation implemented in AMLCS-DA.

5 Observation Operators

Four synthetic observation configurations were tested. In every case, the observation network includes every eligible horizontal grid point. The phrase "fully observed" therefore refers to horizontal coverage; the set of observed state variables differs by experiment.

All-linear experiment. All five state variables are directly observed:

$$\mathcal{H}_{\text{lin}}(\boldsymbol{x}) = \boldsymbol{x}. \tag{15}$$

This is the most favorable configuration for LETKF and serves as a baseline for checking the score filter in a standard DA setting.

All-arctangent experiment. A componentwise arctangent operator is applied to all observed variables. With the climatological mean $\boldsymbol{\mu}_c$, climatological standard deviation $\boldsymbol{\sigma}_c$, and scale factor $s = 1$, the implemented operator is

$$\mathcal{H}_{\text{arctan}}(\boldsymbol{x}) = \arctan\left(s \frac{\boldsymbol{x} - \boldsymbol{\mu}_c}{\boldsymbol{\sigma}_c} \right). \tag{16}$$

This controlled experiment makes the likelihood nonlinear for every variable while avoiding changes to the model dynamics or observation locations.

Wind-direction/wind-speed experiment. Temperature, humidity, and surface pressure are directly observed. Zonal and meridional wind are not observed separately. Instead, the observation vector contains wind direction and wind speed:

$$\mathcal{H}_{\text{WDG}}(u, v) = \text{atan2}(u, v), \tag{17}$$

$$\mathcal{H}_{\text{WSG}}(u, v) = \sqrt{u^2 + v^2}. \tag{18}$$

Directional innovations are wrapped to $[-\pi, \pi]$. This experiment introduces a physically meaningful nonlinear relationship between two state variables. It also resembles real atmospheric observing systems more closely than applying the same artificial transform independently to every variable. Wind-direction error is known to depend strongly on wind speed, especially for weak winds [5]; the present experiment is still idealized because it uses a fixed observation-error model.

Pressure-only experiment. Only surface pressure is directly and linearly observed:

$$\mathcal{H}_P(\boldsymbol{x}) = \text{PSG1}. \tag{19}$$

This older-style DA experiment tests whether a filter can use the atmospheric relationships represented by the ensemble to improve the larger state from a single observed field. Surface pressure is the principal diagnostic for this case.

Table 2: Observation structure and scientific focus of the four experiments.

Experiment	Observations	Primary plotted variables
All linear	Direct UG1, VG1, TG1, TRG1, PSG1	TRG1
All arctangent	Arctangent-transformed UG1, VG1, TG1, TRG1, PSG1	TRG1
WDG/WSG/TPH	Nonlinear WDG1 and WSG1; direct TG1, TRG1, PSG1	UG1 and VG1
Pressure only	Direct PSG1 only	PSG1

The observation-error standard deviations used in the completed experiments are listed in Table 3. For direct observations, the linear-error calculation took 1% of the model-state dynamical range. For each multilevel variable, the maximum-minus-minimum range was calculated separately at each level over all remaining dimensions and then averaged over levels; for surface pressure, 1% of the full-field range was used. The all-arctangent errors were specified directly in the transformed, nondimensional observation space. The direct TG1, TRG1, and PSG1 observations in the WDG/WSG/TPH case use the same errors as the all-linear case, and the pressure-only case uses the same direct PSG1 error.

Table 3: Observation-error standard deviations used in the experiments.

Experiment	Error standard deviations
All linear	UG1: $0.42\,\mathrm{m\,s^{-1}}$; VG1: $0.18\,\mathrm{m\,s^{-1}}$; TG1: $0.46\,\mathrm{K}$; TRG1: $0.05\,\mathrm{g\,kg^{-1}}$; PSG1: $0.006\ \log(p_s/P_0)$
All arctangent	UG1: 0.025; VG1: 0.050; TG1: 0.010; TRG1: 0.010; PSG1: 0.005 (all in transformed observation space)
WDG/WSG/TPH	WDG1: $0.2\,\mathrm{rad}$; WSG1: $1.0\,\mathrm{m\,s^{-1}}$; TG1: $0.46\,\mathrm{K}$; TRG1: $0.05\,\mathrm{g\,kg^{-1}}$; PSG1: $0.006\ \log(p_s/P_0)$
Pressure only	PSG1: $0.006\ \log(p_s/P_0)$

6 Experimental Configuration and Parameter Selection

All experiments used $N_e = 80$ ensemble members and 30 assimilation cycles. Observations were assimilated every two model days, producing a 60-day assimilation window. Eligible variables were observed at 100% of horizontal grid points, corresponding to an AMLCS observation spacing of $s = 1$. The same truth and background ensembles were used when comparing LETKF, EnSF, and the no-data-assimilation (NoDA) baseline.

Localization and multiplicative inflation were tuned separately for LETKF in each experiment. The inflation set was

$$\rho_{\mathrm{LETKF}} \in \{1.00, 1.15, 1.30, 1.45, 1.60\}, \tag{20}$$

and the localization-radius set was

$$r \in \{1, 2, 3, 4\}. \tag{21}$$

The Cartesian product produced 20 LETKF parameter combinations per experiment. The selected combination was the run with the lowest analysis error for the greatest number of state variables. EnSF does not use a localization radius. Its tested inflation values were

$$\rho_{\mathrm{EnSF}} \in \{0.6, 0.8, 1.0, 1.2, 1.4\}. \tag{22}$$

An inflation value of 1.0 was selected for EnSF in all four experiments. Table 4 gives the selected settings. In the compact run-directory names, the decimal point is removed, so `i115` denotes an inflation factor of 1.15.

Table 4: Selected inflation and localization settings.

Experiment	LETKF inflation	LETKF radius	EnSF inflation
All linear	1.00	3	1.00
All arctangent	1.00	3	1.00
WDG/WSG/TPH	1.15	2	1.00
Pressure only	1.30	2	1.00

7 Gaussianity and Evaluation Metrics

The principal verification metric is the root-mean-square error (RMSE) of the ensemble-mean state relative to the synthetic truth. For variable v, level ℓ, and cycle k, the archived plotting script calculates

$$\mathrm{RMSE}_{v,\ell}(k) = \left[\frac{1}{N_\phi N_\lambda} \sum_{i=1}^{N_\phi} \sum_{j=1}^{N_\lambda} \left(\overline{x}_{v,\ell,i,j}(k) - x^{\mathrm{true}}_{v,\ell,i,j}(k) \right)^2 \right]^{1/2}, \tag{23}$$

where $N_\phi = 32$ and $N_\lambda = 64$. This is an unweighted arithmetic mean over the latitude–longitude grid; no cosine-latitude weighting is applied. The same calculation is performed separately for the analysis mean, background mean, and NoDA field. The plotted level-averaged diagnostic is then the arithmetic mean of the independently calculated level RMSE values,

$$\mathrm{RMSE}_{v,\mathrm{levavg}}(k) = \frac{1}{L_v} \sum_{\ell \in \mathcal{L}_v} \mathrm{RMSE}_{v,\ell}(k), \tag{24}$$

not a single RMSE computed after pooling every grid point and level. The level set contains all eight levels for UG1, VG1, and TG1, levels 2–7 for TRG1, and the single surface level for PSG1. The plots show analysis RMSE with solid lines, background RMSE with dashed lines, and NoDA error with a black line. EnSF is blue and LETKF is orange. Except for the pressure-only focus plot, the vertical axis is logarithmic so that differences among the low-error analysis curves remain visible.

Two additional EnSF diagnostics complement the domain-averaged RMSE. The spatial maps show the pointwise absolute ensemble-mean error at the flagship level before any assimilation and after the first EnSF analysis. The pre-assimilation field is taken from `free_run_0.nc`, and both maps in each figure use the same logarithmic color scale. The reverse-SDE trajectory diagnostic records all 80 ensemble members through the 1,000 pseudotime steps in physical units. It shows the spatial-mean tracked state together with grid index $(24, 36)$, which was selected during preprocessing as the location with the largest analysis increment. These diagnostics illustrate where and how EnSF changes the state; the RMSE curves remain the primary basis for the comparison with LETKF.

Humidity was placed at the forefront of the all-linear and all-arctangent experiments because its ensemble distribution was the least Gaussian of the five state variables. A Shapiro-Wilk test [11] was applied at each grid point using the 80 background-ensemble members as the sample. The gridpoint p-values were averaged without spatial weighting to obtain one value per model level, after which the valid level means were averaged to obtain the final value for each variable. Numerically constant ensemble samples were treated as undefined and excluded; this removes the constant upper humidity layers. The diagnostic calculated for this project is reproduced in Table 5. Humidity has the smallest level-averaged value (0.2509) and the smallest values at every level on which it is defined. The other multilevel variables have level-averaged values between 0.4249 and 0.4354, and surface pressure has a value of 0.4396. These results identify humidity as the strongest departure from the approximately Gaussian behavior favored by ensemble Kalman methods. Because the table averages many local hypothesis-test results, it is used as a comparative diagnostic rather than as a single field-wide hypothesis test.

Table 5: Unweighted grid-mean Shapiro-Wilk p-value diagnostic by SPEEDY level. Humidity is defined only on levels 2–7; constant upper layers are excluded. The final row is the unweighted mean over valid levels.

Level	UG1	VG1	TG1	TRG1	PSG1
0	0.4637	0.4257	0.4392	–	0.4396
1	0.3990	0.4396	0.5019	–	–
2	0.4079	0.4298	0.4405	0.2473	–
3	0.4191	0.4163	0.4812	0.2471	–
4	0.4338	0.4337	0.4215	0.1985	–
5	0.4386	0.4370	0.3927	0.2495	–
6	0.4436	0.4490	0.3842	0.2716	–
7	0.3937	0.3950	0.4222	0.2915	–
Level average	0.4249	0.4282	0.4354	0.2509	0.4396

Results

1 All-Linear Observations

Analysis and Background Performance

The all-linear experiment provides the expected strongest setting for LETKF. With direct observations of every state variable at every horizontal grid point, LETKF produced lower analysis error than EnSF for zonal wind, meridional wind, temperature, and surface pressure. Its background curves also benefited from the more accurate analyses carried into the next two-day model forecast. Both assimilation methods substantially improved upon NoDA, showing that both filters produced useful analyses in the full SPEEDY state even where LETKF achieved the lower error.

Humidity is the notable exception. As shown in Figure 1, the two humidity analysis curves converge to similar values, with EnSF slightly lower over the final cycles. This result is consistent with the Gaussianity diagnostic in Table 5. LETKF is especially effective for a linear observation model and an approximately Gaussian ensemble, whereas humidity was the least Gaussian variable tested. Its distribution therefore offers a plausible reason that the score filter remained competitive even with a linear observation operator. This is not an overall score-filter advantage in the linear experiment; instead, it indicates that distributional structure can matter within an otherwise favorable setting for LETKF.

Figure 2 combines the remaining four variables and establishes the broader LETKF advantage against which the humidity result should be interpreted. Figure 3 provides a complementary spatial diagnostic for EnSF. The first EnSF analysis reduces the widespread level-7 humidity error relative to the pre-assimilation state, consistent with the rapid RMSE decrease at the beginning of the flagship time series.

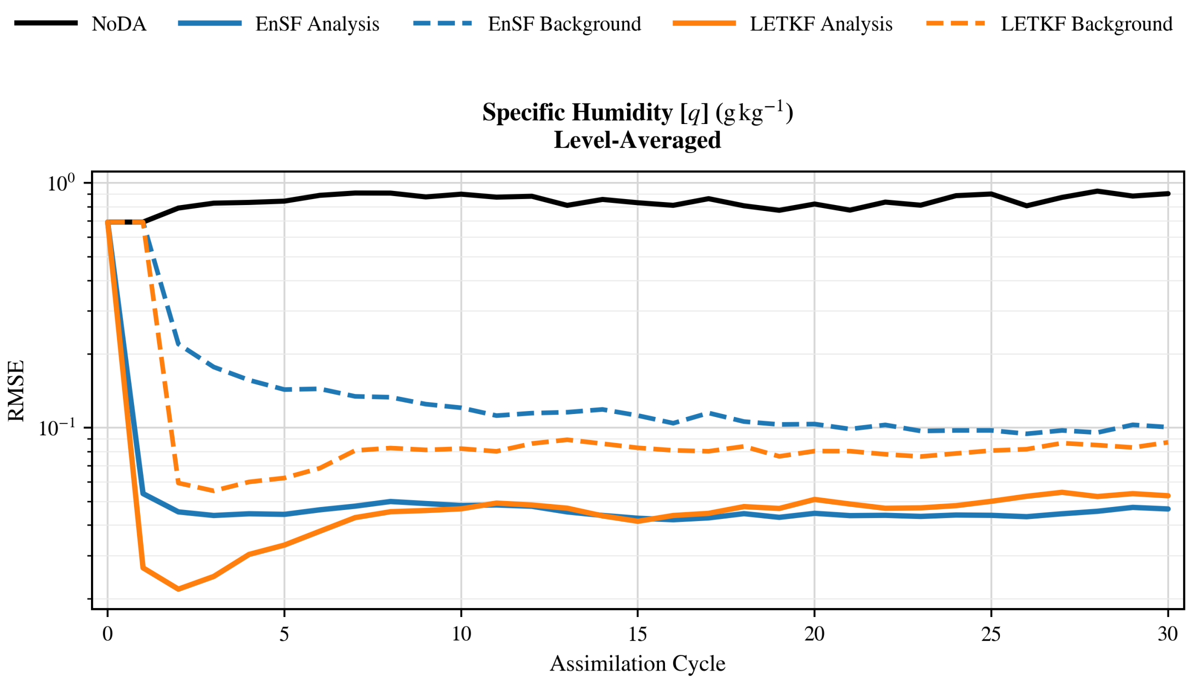

Figure 1: All-linear experiment: level-averaged specific-humidity RMSE on a logarithmic vertical axis. Humidity is the flagship variable because it showed the strongest non-Gaussianity. EnSF is blue, LETKF is orange, and NoDA is black; solid and dashed curves denote analysis and background, respectively. The selected LETKF uses inflation 1.00 and localization radius 3, while EnSF uses inflation 1.00.

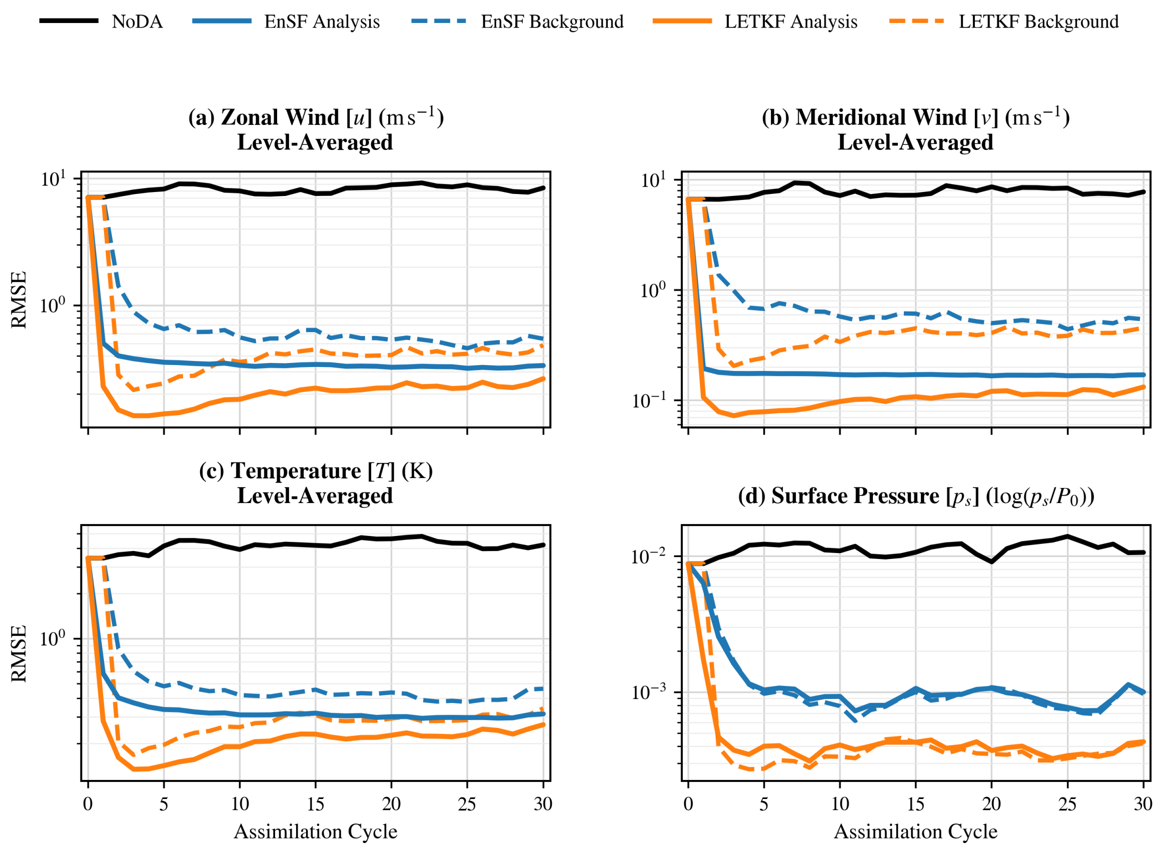

Figure 2: All-linear experiment: logarithmic RMSE for the four secondary variables. Panels show level-averaged zonal wind, meridional wind, and temperature, and level-0 surface pressure. LETKF produces the lower analysis error in each panel.

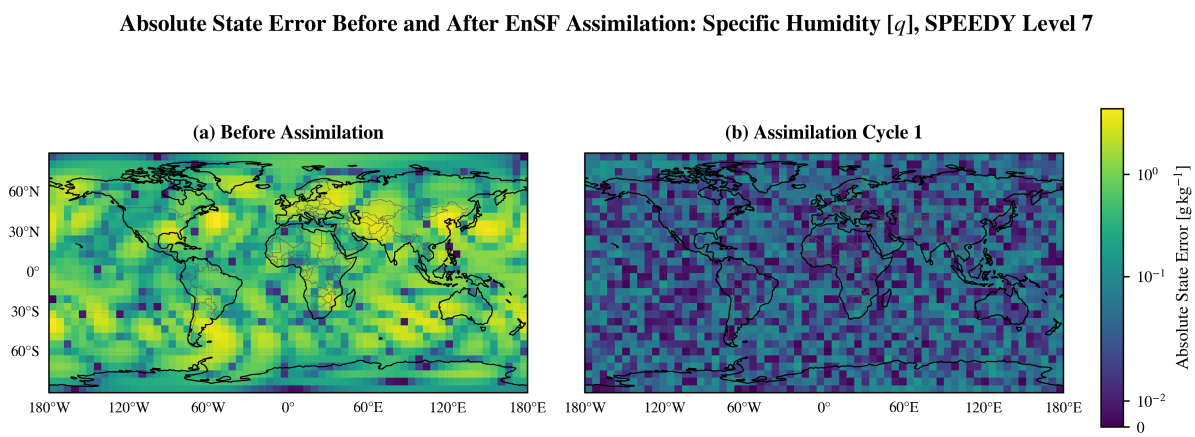

Figure 3: All-linear experiment: absolute level-7 specific-humidity error before any assimilation and after the first EnSF analysis. Both maps use the same logarithmic color scale, so the change in color directly shows the spatial reduction in analysis error.

2 All-Arctangent Observations

Analysis and Background Performance

Applying the arctangent operator to every observed variable changes the result decisively. EnSF achieved lower analysis error for all five variables, including humidity in Figure 4 and the four secondary fields in Figure 5. The score filter evaluates the nonlinear likelihood gradient and its exact chain-rule factor throughout the reverse diffusion. This permits a nonlinear deformation of the analysis sample distribution rather than restricting the analysis to an affine ensemble transform.

LETKF nevertheless performed well and remained substantially below NoDA. The arctangent operator is smooth and monotone, so the background ensemble can still provide useful observation-space means, perturbations, and cross-covariances. LETKF's improvement over NoDA shows that it remains viable in this setting, while the consistent EnSF advantage across all variables shows the benefit of the score update when the likelihood is nonlinear throughout the state.

The humidity comparison remains the most important panel because the experiment combines a nonlinear observation operator with the state variable that already showed the strongest non-Gaussianity. Its outcome supports the central motivation for score-based DA more directly than the linear humidity tie.

The spatial maps in Figure 6 show that the first EnSF analysis reduces level-7 humidity error across most of the globe. The sampled paths in Figure 7 expose the corresponding reverse-SDE mechanism. Starting from the noisy state at pseudotime $t = 1$, the ensemble is progressively concentrated toward its analysis distribution as t approaches zero. The grid point with the largest analysis increment is shown alongside the spatial-mean state to illustrate both a domain-wide average and a location where the update is especially strong.

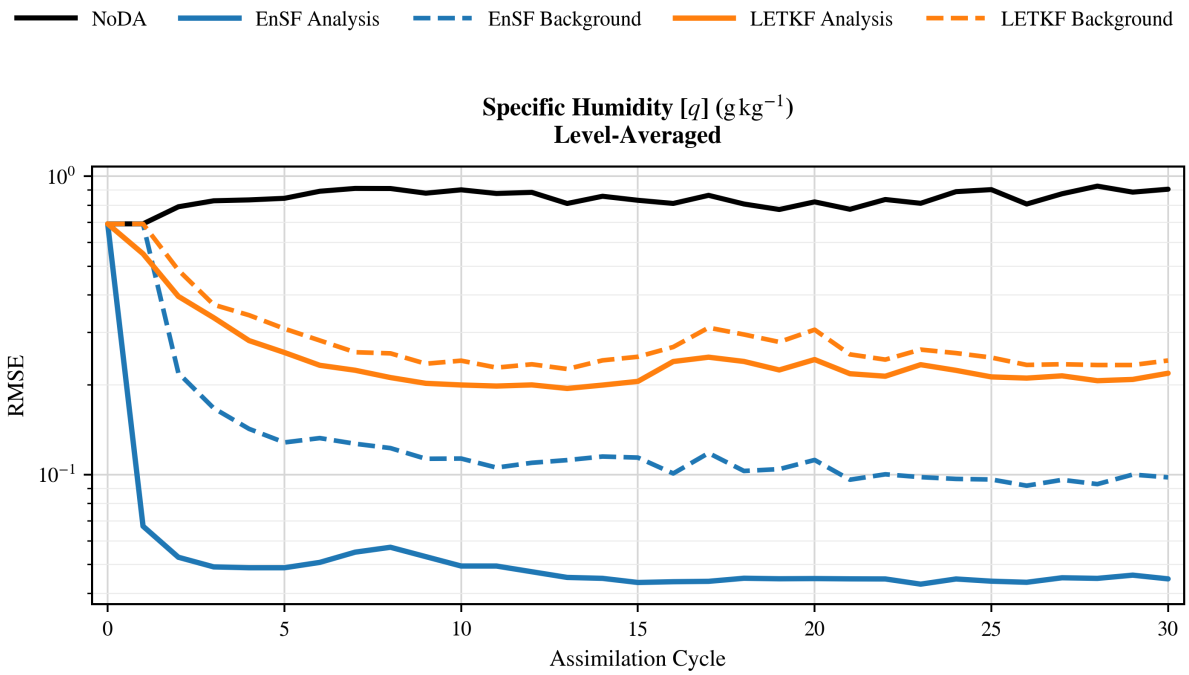

Figure 4: All-arctangent experiment: level-averaged specific-humidity RMSE on a logarithmic vertical axis. EnSF produces substantially lower analysis and background errors than LETKF for the least Gaussian state variable.

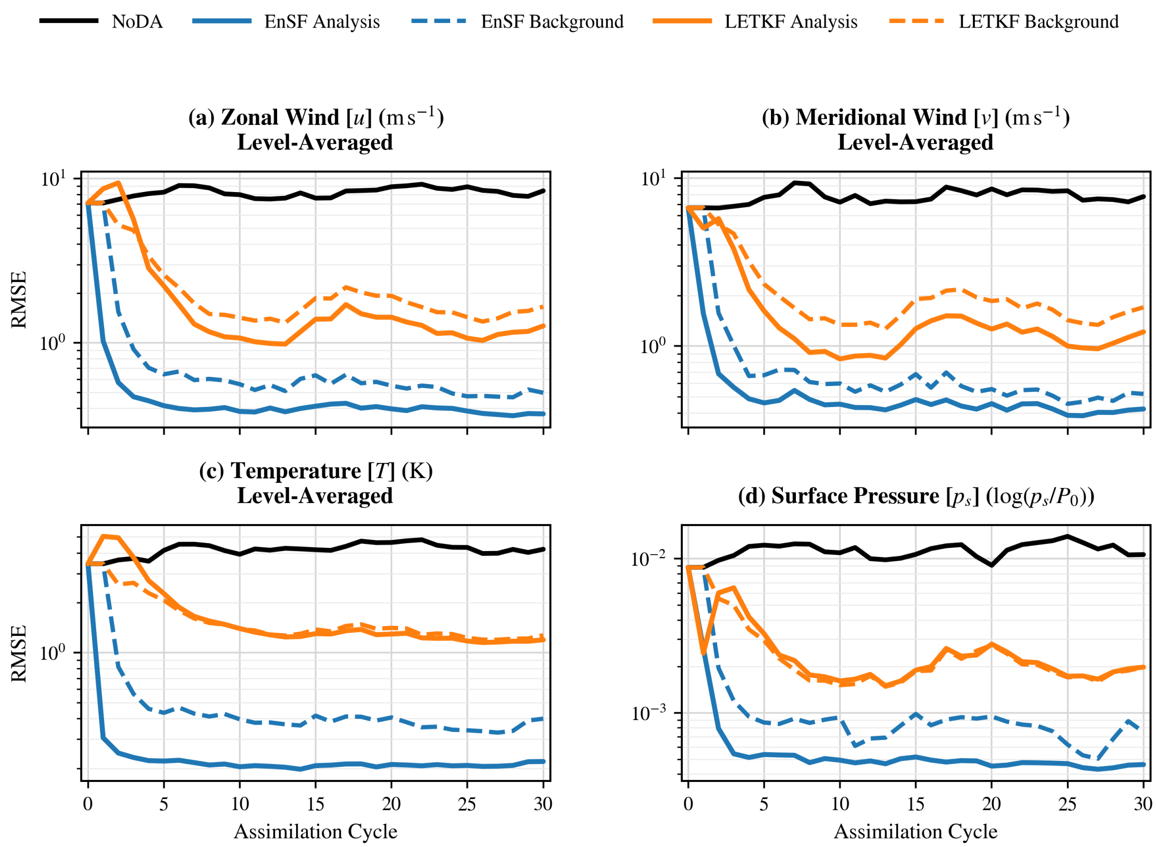

Figure 5: All-arctangent experiment: logarithmic RMSE for the four secondary variables. Panels show level-averaged zonal wind, meridional wind, and temperature, and level-0 surface pressure. EnSF has the lower analysis error in all four panels.

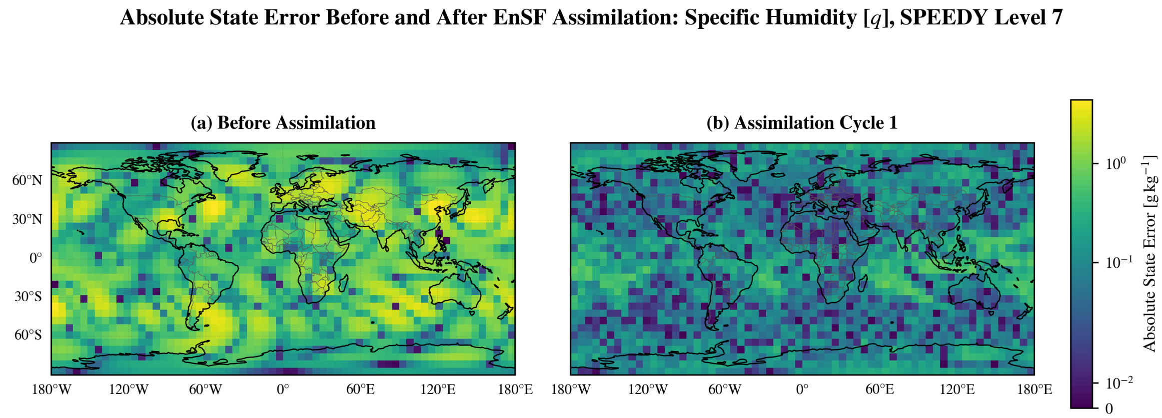

Figure 6: All-arctangent experiment: absolute level-7 specific-humidity error before any assimilation and after the first EnSF analysis. The common logarithmic color scale shows a widespread reduction in spatial error.

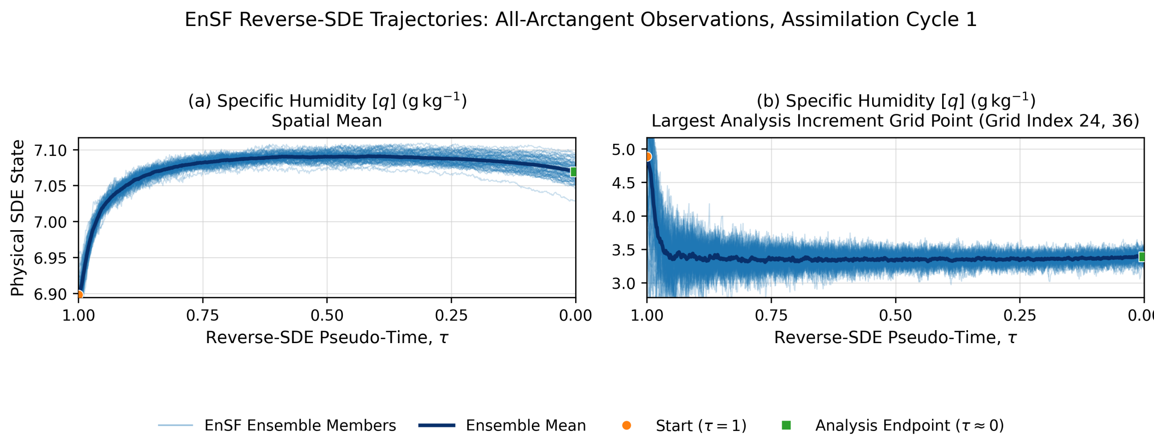

Figure 7: All-arctangent experiment: EnSF reverse-SDE trajectories for specific humidity during the first assimilation cycle. The left panel shows the spatial-mean tracked state; the right shows grid index $(24, 36)$, selected for the largest analysis increment. Thin curves are the 80 ensemble members, the dark curve is their mean, and markers identify the start at $\tau = 1$ and the analysis endpoint near $\tau = 0$. Values are in physical units.

3 Nonlinear Wind Observations (WDG/WSG/TPH)

Wind Performance

The WDG/WSG/TPH experiment replaces direct observations of u and v with the coupled nonlinear observations in Equation (18). Zonal and meridional wind are therefore the principal diagnostics and are displayed at full size in Figure 8. LETKF produced the lowest error for both wind components. It also outperformed EnSF for temperature, humidity, and surface pressure in Figure 9.

At first glance, the wind result is surprising because direction and speed form a nonlinear observation operator for the two wind components. Two complementary mechanisms may explain it. First, temperature, pressure, and humidity remain directly and linearly observed at every horizontal grid point. The localized LETKF ensemble contains cross-covariances between these fields and the winds, allowing the thermodynamic observations to contribute to the wind analysis in addition to the WDG and WSG observations. The original EnSF used here evaluates the exact wind-direction and wind-speed likelihood gradients, but it does not have an equivalent localized multivariate regression step. The cross-variable pathway therefore provides LETKF with information that is not used in the same way by EnSF.

Second, although the transformation from (u, v) to wind speed and direction is nonlinear, the two quantities together retain nearly all information about the wind vector. Away from calm winds and the angular branch cut, the mapping is locally smooth and invertible. It is therefore less ambiguous than a strongly saturating nonlinear observation applied independently to each variable, as in the all-arctangent experiment. With dense observations and a sufficiently compact background ensemble, LETKF's local linear ensemble relationships may approximate this transformation well enough that the nonlinear operator alone does not create an EnSF advantage.

Figure 10 shows that EnSF nevertheless reduces widespread level-7 errors in both wind components during the first analysis. Thus, its higher RMSE over most of the assimilation window is not simply a failure to respond to the nonlinear wind observations. The current diagnostics cannot separate the two mechanisms above. A WDG/WSG-only ablation, with the direct temperature, humidity, and pressure observations removed, would test the cross-covariance explanation. Stratifying wind errors by speed and proximity to the directional branch cut would test whether the remaining nonlinearity is concentrated in the regimes where the speed-direction mapping is least regular.

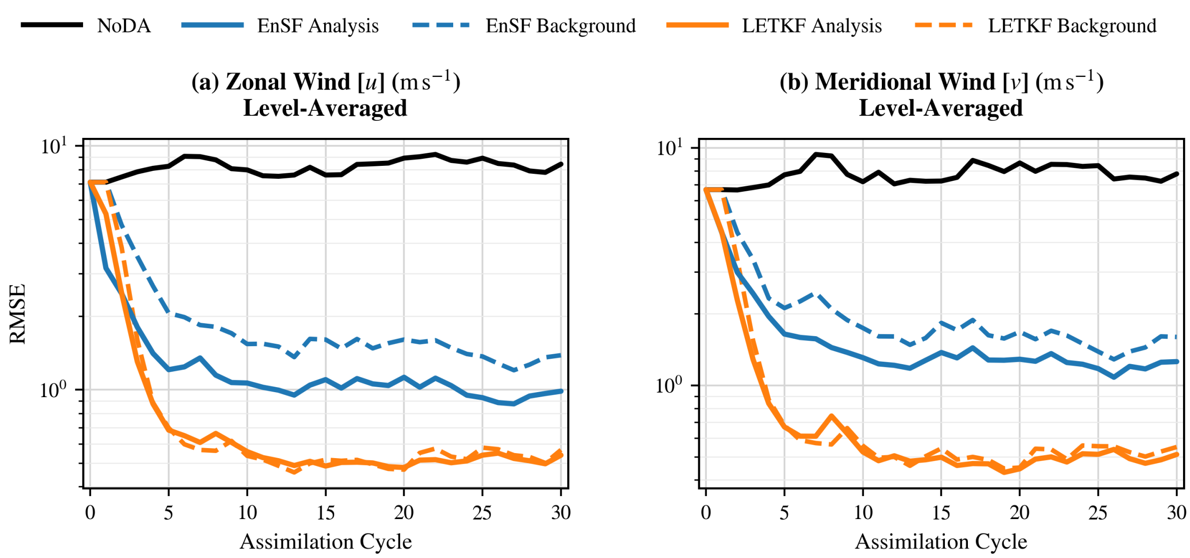

Figure 8: WDG/WSG/TPH experiment: logarithmic, level-averaged RMSE for zonal wind (left) and meridional wind (right). Both components are observed indirectly through wind direction and wind speed. LETKF has lower analysis and background errors for both flagship variables.

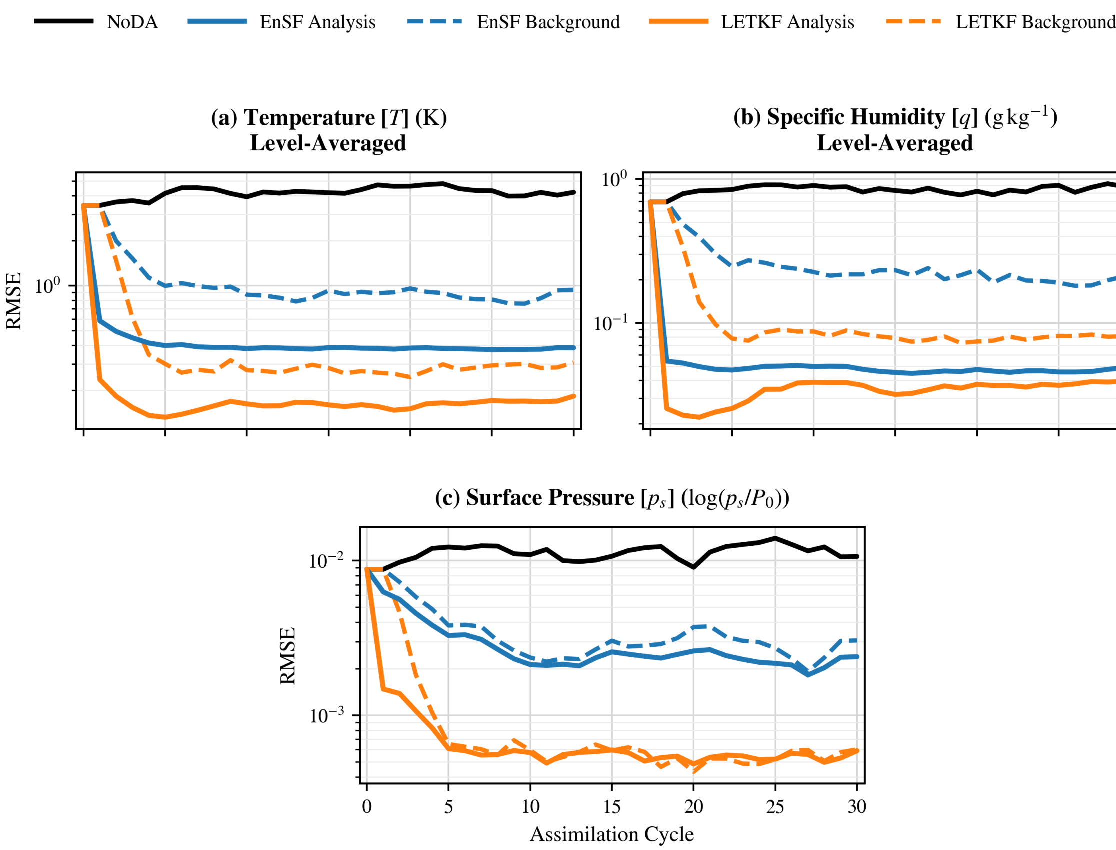

Figure 9: WDG/WSG/TPH experiment: logarithmic RMSE for the directly observed thermodynamic variables. Panels show level-averaged temperature and specific humidity and level-0 surface pressure. LETKF has lower analysis error in all three panels.

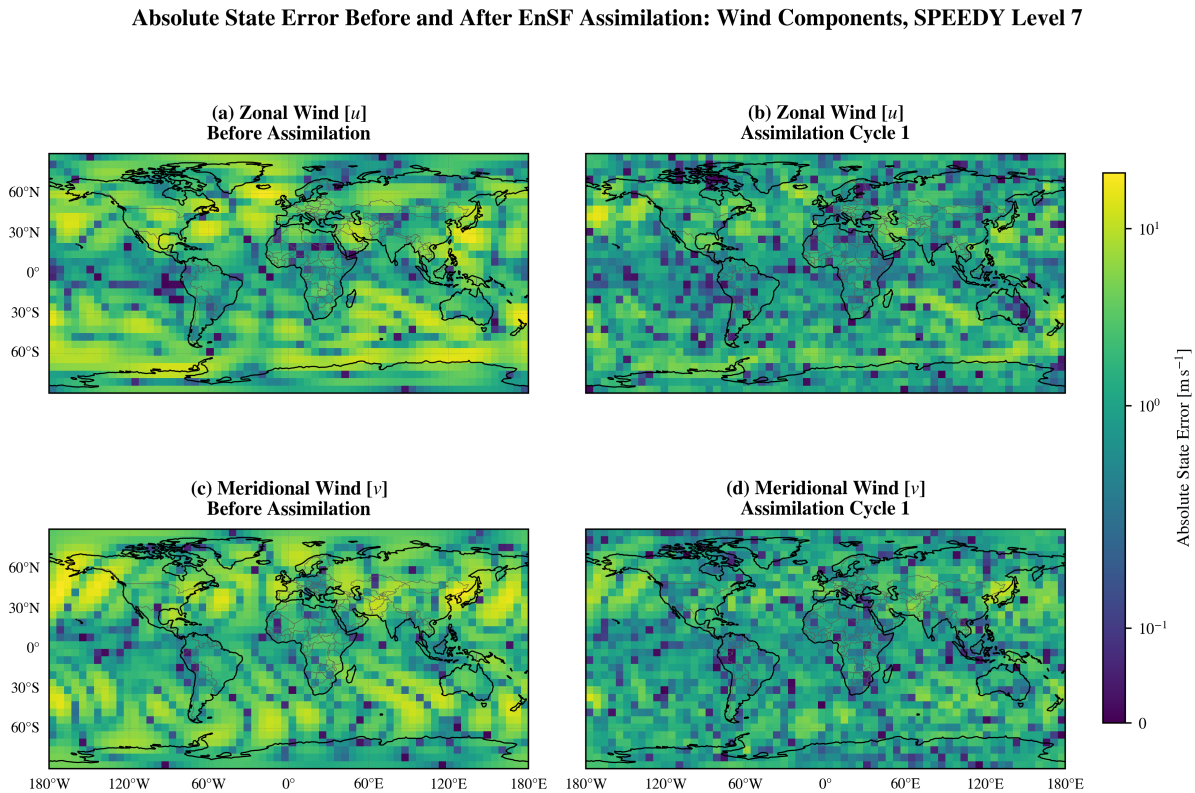

Figure 10: WDG/WSG/TPH experiment: absolute level-7 zonal- and meridional-wind errors before any assimilation (left column) and after the first EnSF analysis (right column). All four panels use a common logarithmic color scale.

4 Pressure-Only Observations

Surface-Pressure Performance

Surface pressure is the only directly observed field in this experiment, so Figure 11 is the focus and is shown with a linear vertical axis. LETKF produced clearly lower surface-pressure error. EnSF still improved meaningfully upon NoDA, demonstrating that its direct pressure update extracted useful information from the observations, but it did not match the localized covariance-based LETKF analysis.

This case emphasizes a different limitation than the all-arctangent experiment. The pressure observation operator is linear, but the observation network is partial in state-variable space. LETKF uses ensemble-estimated cross-covariances to spread the observed pressure information through each local atmospheric state. In the original score-filter formulation, the likelihood gradient acts directly on the observed component and does not provide an equivalent analysis-time regression to unobserved components. The result is therefore consistent with the mechanisms of the two methods.

Figure 12 documents the four unobserved state variables and their indirect evolution through the forecast model. Figure 13 confirms that the first EnSF update produces a spatially broad reduction in surface-pressure error even though its subsequent RMSE remains above LETKF. Conclusions for this experiment are based primarily on surface pressure, since it is the only observed quantity.

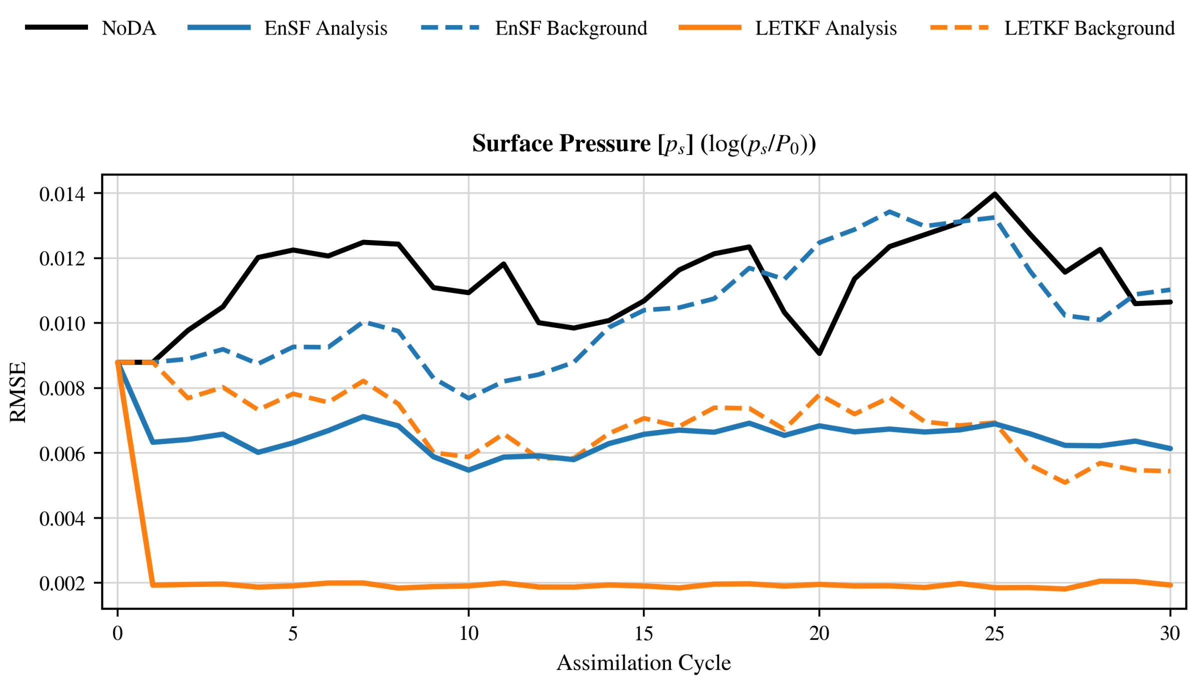

Figure 11: Pressure-only experiment: level-0 surface-pressure RMSE on a linear vertical axis. LETKF is clearly more accurate, while EnSF still improves upon NoDA.

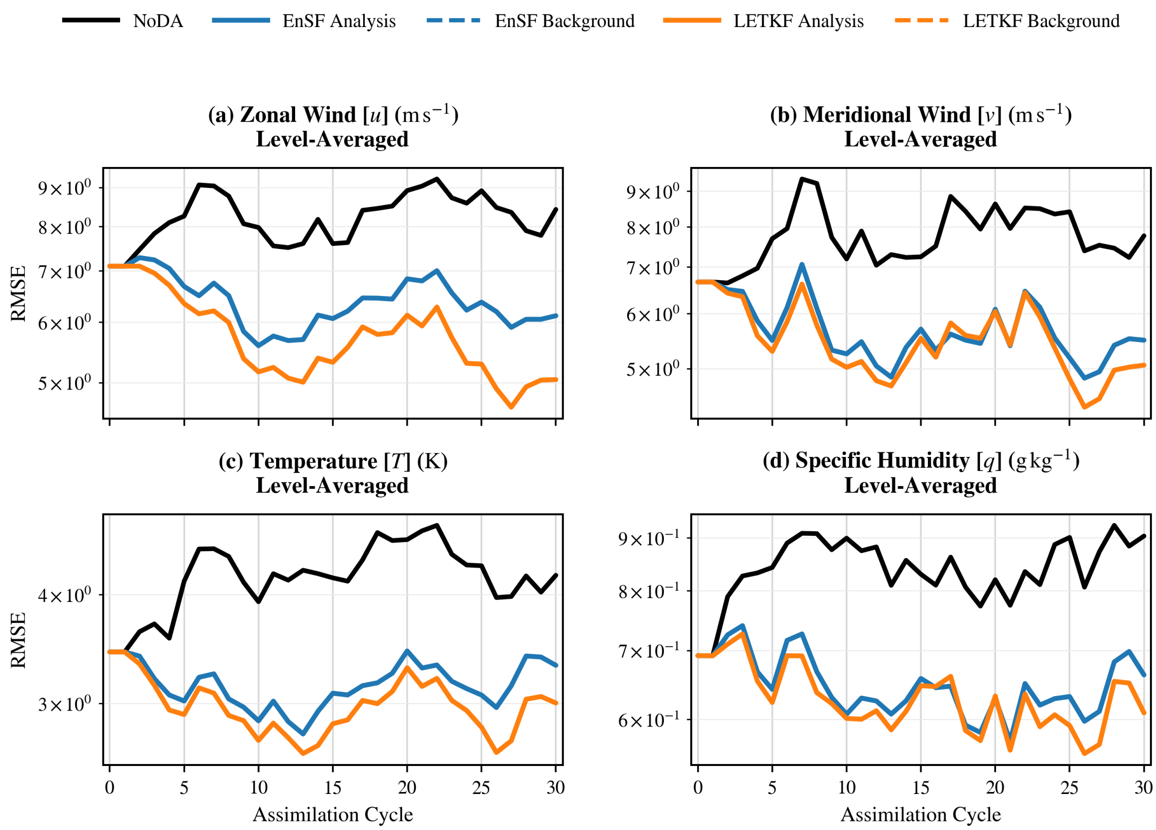

Figure 12: Pressure-only experiment: logarithmic RMSE for the four unobserved secondary variables. Panels show level-averaged zonal wind, meridional wind, temperature, and specific humidity. These fields can respond only indirectly through later model evolution in the original EnSF configuration.

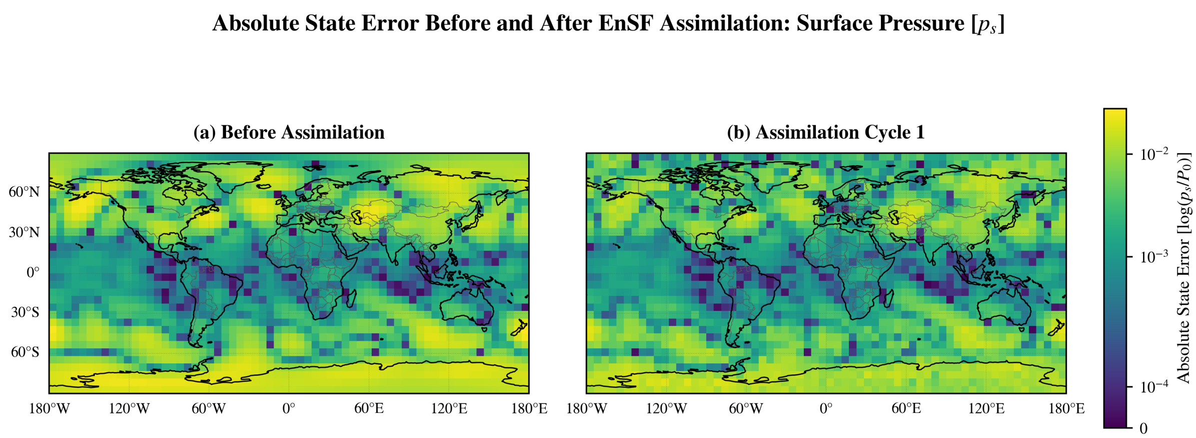

Figure 13: Pressure-only experiment: absolute level-0 surface-pressure error before any assimilation and after the first EnSF analysis. The two maps use a common logarithmic color scale.

5 Cross-Experiment Comparison

Table 6 summarizes the qualitative result of each experiment. No single method dominated every observing configuration. LETKF was strongest when the observation system was linear and dense, and it retained an advantage under the coupled WDG/WSG operator because it could use localized cross-variable information from the directly observed thermodynamic fields. It was also clearly stronger when only pressure was observed. EnSF was strongest when the arctangent operator made the entire observation system nonlinear, and it reached parity with LETKF for the most non-Gaussian variable in the all-linear experiment.

Table 6: Qualitative comparison of the selected LETKF and EnSF runs.

Experiment	Lower analysis error	Main interpretation
All linear	LETKF for UG1, VG1, TG1, PSG1; approximately even for TRG1	Linear, dense observations favor LETKF; humidity non-Gaussianity narrows the difference.
All arctangent	EnSF for all five variables	A nonlinear likelihood throughout the state favors the score update.
WDG/WSG/TPH	LETKF for all five variables	Local cross-variable coupling and the information-preserving speed-direction pair offset the nonlinear wind operator.
Pressure only	LETKF clearly for PSG1	LETKF spreads pressure information through ensemble covariances; EnSF still improves observed pressure over NoDA.

These results refine the expectation that score-based filtering should prevail whenever an observation operator is nonlinear. The all-arctangent case supports that expectation, but the nonlinear wind case shows that the observation network and multivariate update mechanism can be equally important. The comparison is therefore best understood along two axes: the nonlinearity and distributional structure of the estimation problem, and each method's ability to transfer information among state variables and locations.

Table 7 reports the mean analysis RMSE across the 30-cycle assimilation window for the selected LETKF and EnSF configurations. For each cycle, horizontal RMSE was calculated independently at each retained model level and then averaged across levels. Averaging the resulting series across all cycles summarizes sustained performance rather than emphasizing a single best or final cycle.

Table 7: Mean analysis RMSE across 30 assimilation cycles for the selected LETKF and EnSF configurations.

Experiment	Variable	Units	LETKF analysis	EnSF analysis
All linear	UG1	$\mathrm{m\,s^{-1}}$	0.2031	0.3447
	VG1	$\mathrm{m\,s^{-1}}$	0.1033	0.1713
	TG1	K	0.2094	0.3247
	TRG1	$\mathrm{g\,kg^{-1}}$	0.04417	0.04546
	PSG1	$\log(p_s/P_0)$	4.329×10^{-4}	1.191×10^{-3}
All arctangent	UG1	$\mathrm{m\,s^{-1}}$	1.995	0.4248
	VG1	$\mathrm{m\,s^{-1}}$	1.556	0.4917
	TG1	K	1.723	0.2154
	TRG1	$\mathrm{g\,kg^{-1}}$	0.2410	0.04761
	PSG1	$\log(p_s/P_0)$	2.404×10^{-3}	5.654×10^{-4}
WDG/WSG/TPH	UG1	$\mathrm{m\,s^{-1}}$	0.7961	1.203
	VG1	$\mathrm{m\,s^{-1}}$	0.7439	1.526
	TG1	K	0.1624	0.3965
	TRG1	$\mathrm{g\,kg^{-1}}$	0.03425	0.04790
	PSG1	$\log(p_s/P_0)$	6.354×10^{-4}	2.771×10^{-3}
Pressure only	UG1	$\mathrm{m\,s^{-1}}$	5.711	6.380
	VG1	$\mathrm{m\,s^{-1}}$	5.469	5.669
	TG1	K	2.970	3.148
	TRG1	$\mathrm{g\,kg^{-1}}$	0.6272	0.6467
	PSG1	$\log(p_s/P_0)$	1.921×10^{-3}	6.430×10^{-3}

Summary and Discussion

This project compared an ensemble score filter with the Local Ensemble Transform Kalman Filter in the SPEEDY intermediate-complexity atmospheric general circulation model. The central question was whether score-filter benefits demonstrated in low-dimensional and otherwise idealized benchmarks would persist in a multivariable, vertically resolved global atmosphere that remains computationally manageable. Both methods were implemented within the same AMLCS-DA forecast-analysis framework, used 80 ensemble members, assimilated observations at two-day intervals for 30 cycles, and were compared with the same NoDA reference.

The all-linear experiment behaved largely as expected. LETKF produced lower analysis error for wind, temperature, and surface pressure. For specific humidity, however, the two methods were essentially even after 30 cycles. The Shapiro-Wilk diagnostic identified humidity as the least Gaussian state variable, providing a plausible explanation for why the score filter was most competitive for humidity even under a linear observation operator.

The all-arctangent experiment provided the clearest score-filter result. EnSF produced lower error for every state variable, while LETKF still substantially improved upon NoDA. This demonstrates the central strength of the score formulation: it evaluates a nonlinear likelihood gradient throughout a reverse diffusion rather than applying only an affine ensemble analysis. The result also shows that the score-filter advantage can persist beyond simplified benchmark systems in a global primitive-equation atmosphere.

The physically motivated nonlinear wind experiment produced a more nuanced result. LETKF outperformed EnSF for both wind components and for the three directly observed thermodynamic variables. The likely explanation is not a single effect. LETKF could exploit localized cross-covariances connecting wind to the linearly observed temperature, humidity, and pressure fields. At the same time, wind speed and direction together almost uniquely determine the wind vector and are locally smooth away from calm winds and the angular branch cut. The nonlinear transformation may therefore remain manageable for LETKF over most of the background ensemble. The first-cycle maps confirm that EnSF makes a substantial spatial correction, but they do not identify which of these two mechanisms controls the longer-term method difference. A WDG/WSG-only ablation is needed to isolate the contribution of the linearly observed thermodynamic variables.

LETKF was also clearly stronger in the pressure-only experiment. EnSF meaningfully improved observed pressure over NoDA, but the original score filter lacks LETKF's direct analysis-time mechanism for spreading pressure increments to unobserved variables. This result is consistent with recent work that augments EnSF with ensemble linear regression for partially observed systems [14].

Several limitations should be considered. First, these observing-system simulation experiments use a perfect-model framework and 100% horizontal coverage for each eligible observation type. Real observations are spatially and temporally incomplete and include representation, calibration, and operator errors not captured here. Second, the results use one ensemble size, one assimilation interval, and one truth trajectory. Multiple random seeds and truth periods are needed to quantify sampling variability. Third, LETKF was tuned over both localization and inflation, whereas EnSF was tuned only over inflation. Although this reflects the actual parameterization of the two methods, the unequal searches should be acknowledged. Fourth, level-averaged RMSE compresses spatial and vertical information and does not measure ensemble calibration. Rank histograms, spread-skill ratios, continuous ranked probability scores, and spatial error maps would provide a more complete comparison. Finally, computational cost was not a primary metric. Reverse diffusion requires many pseudotime evaluations at each analysis cycle, whereas LETKF performs many local matrix operations. Timing and scaling tests are necessary before drawing conclusions about operational feasibility.

The most direct next step is to implement the two-step EnSF with regression in AMLCS-DA and repeat the pressure-only and WDG/WSG experiments. That extension would preserve the nonlinear observation-space score update while using ensemble cross-covariances to update unobserved variables. Additional work should reduce the observation density, vary the ensemble size and assimilation interval, test multiple truth trajectories, and examine observation errors that depend on atmospheric conditions. In particular, a wind-direction error model that increases at low wind speed would make the WDG/WSG case more realistic.

Overall, the project shows that the ensemble score filter can provide a clear advantage in a nonlinear global-atmosphere experiment, especially for strongly non-Gaussian variables and nonlinear likelihoods. It also shows why LETKF remains a demanding benchmark: localization and multivariate ensemble covariances make it highly effective when observations are linear, partial, or coupled to other well-observed variables. The comparison therefore supports continued development of score-based DA, particularly hybrid methods that combine nonlinear score updates with the information-transfer strengths of ensemble Kalman filtering.

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